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Duration	8 mins 9 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The problem is that we want to reverse a array in O(N) linear time complexity and we want the algorithm to be in-place as well!
For example: input is [1,2,3,4,5] then the output is [5,4,3,2,1]
Input
5
1 2 3 4 5
Output
5 4 3 2 1

For example:

Input	Result
5 1 2 3 4 5	5 4 3 2 1

Answer: (penalty regime: 0 %)

```
import sys


def reverse_array_in_place(arr: list[int]) -> None:
    left = 0
    right = len(arr) - 1

    # Two-pointer swap
    while left < right:
        arr[left], arr[right] = arr[right], arr[left]
        left += 1
        right -= 1


def solve():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])
    arr = [int(x) for x in tokens[1 : n + 1]]

    reverse_array_in_place(arr)

    print(" ".join(map(str, arr)))


if __name__ == "__main__":
    solve()
```

	Input	Expected	Got	
✓	5 1 2 3 4 5	5 4 3 2 1	5 4 3 2 1	✓
✓	10 0 2 4 6 8 1 3 5 7 9	9 7 5 3 1 8 6 4 2 0	9 7 5 3 1 8 6 4 2 0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Objective:

Develop a Python program to find the k-th maximum value in a given list of integers. The program should handle various edge cases, including lists with duplicate values, empty lists, and invalid values of k. The k-th maximum value refers to the k-th largest distinct element in the list.

Background:

Finding the k-th maximum value in a list is a common problem in computer science, often encountered in fields like data analysis, competitive programming, and software development. This problem requires an understanding of sorting algorithms, data structures, and efficient problem-solving techniques. By solving this problem, one gains insights into how to handle large datasets and optimize performance in practical applications.

Problem Description:

Given a list of integers, the task is to determine the k-th maximum value in the list. The program should meet the following requirements:

1. **Input:**
    - A list of integers, which can contain both positive and negative values.
    - An integer k, representing the position of the maximum value to find.
2. **Output:**
    - The k-th maximum value in the list.
    - If k is greater than the number of distinct elements in the list or if the list is empty, the program should return an appropriate message indicating the error.

Constraints:

- The list may contain duplicate values.
- The value of k should be a positive integer.
- The list may contain up to 10^6 elements, and each element can be as large as 10^9 in magnitude.

Examples:

Consider the following examples for better understanding:

1. **Example 1:**
    - **Input:** list = [3, 1, 5, 4, 2], k = 2
    - **Output:** 4
    - **Explanation:** The distinct elements in the list are [1, 2, 3, 4, 5]. The 2nd maximum value is 4.
2. **Example 2:**
    - **Input:** list = [7, 7, 7, 7, 7], k = 1
    - **Output:** 7
    - **Explanation:** The distinct elements in the list are [7]. The 1st maximum value is 7.
3. **Example 3:**
    - **Input:** list = [2, 1, 2, 1, 2], k = 3
    - **Output:** -1
    - **Explanation:** The distinct elements in the list are [1, 2]. There is no 3rd maximum value.

For example:

Input	Result
5 3 1 5 4 2 2	4
6 7 7 7 7 7 7 1	7
10 2 1 2 1 2 1 2 1 2 1 3	-1

Answer: (penalty regime: 0 %)

```python
import sys


def find_kth_max():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])

    # Extract list elements based on n
    arr = [int(x) for x in tokens[1 : n + 1]]
    k = int(tokens[n + 1])

    # Extract distinct elements and sort in descending order
    distinct_sorted = sorted(set(arr), reverse=True)

    # Check validity of k
    if 1 <= k <= len(distinct_sorted):
        print(distinct_sorted[k - 1])
    else:
        print(-1)


if __name__ == "__main__":
    find_kth_max()
```

	Input	Expected	Got	
✓	5 3 1 5 4 2 2	4	4	✓
✓	6 7 7 7 7 7 7 1	7	7	✓
✓	10 2 1 2 1 2 1 2 1 2 1 3	-1	-1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Write a Python program to sort a list of elements using the merge sort algorithm.

For example:

Input	Result
5 6 5 4 3 8	3 4 5 6 8

Answer: (penalty regime: 0 %)

```
import sys


def merge(left: list[int], right: list[int]) -> list[int]:
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result.extend(left[i:])
    result.extend(right[j:])
    return result


def merge_sort(arr: list[int]) -> list[int]:
    if len(arr) <= 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    return merge(left, right)


def solve():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])
    arr = [int(x) for x in tokens[1 : n + 1]]

    sorted_arr = merge_sort(arr)
    print(" ".join(map(str, sorted_arr)))


if __name__ == "__main__":
    solve()
```

	Input	Expected	Got	
✓	5 6 5 4 3 8	3 4 5 6 8	3 4 5 6 8	✓
✓	9 14 46 43 27 57 41 45 21 70	14 21 27 41 43 45 46 57 70	14 21 27 41 43 45 46 57 70	✓
✓	4 86 43 23 49	23 43 49 86	23 43 49 86	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums` sorted in **non-decreasing** order, return *an array of **the squares of each number** sorted in non-decreasing order*.

Example 1:

```
Input: nums = [-4,-1,0,3,10]
Output: [0,1,9,16,100]
Explanation: After squaring, the array becomes [16,1,0,9,100].
After sorting, it becomes [0,1,9,16,100].
```

Example 2:

```
Input: nums = [-7,-3,2,3,11]
Output: [4,9,9,49,121]
```

Constraints:

- `1 <= nums.length <= 10^4`
- `-10^4 <= nums[i] <= 10^4`
- `nums` is sorted in **non-decreasing** order.

For example:

Test	Result
`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`

Answer: (penalty regime: 0 %)

Reset answer

```
def sortedSquares(nums: list[int]) -> list[int]:
    n = len(nums)
    result = [0] * n
    left = 0
    right = n - 1
    position = n - 1

    while left <= right:
        left_square = nums[left] * nums[left]
        right_square = nums[right] * nums[right]

        if left_square > right_square:
            result[position] = left_square
            left += 1
        else:
            result[position] = right_square
            right -= 1

        position -= 1

    return result
```

	Test	Expected	Got	
✓	`print(sortedSquares([-4,-1,0,3,10]))`	`[0, 1, 9, 16, 100]`	`[0, 1, 9, 16, 100]`	✓
✓	`print(sortedSquares([-7,-3,2,3,11]))`	`[4, 9, 9, 49, 121]`	`[4, 9, 9, 49, 121]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array of integers `arr`, replace each element with its rank.

The rank represents how large the element is. The rank has the following rules:

- Rank is an integer starting from 1.
- The larger the element, the larger the rank. If two elements are equal, their rank must be the same.
- Rank should be as small as possible.

Example 1:

```
Input: arr = [40,10,20,30]
Output: [4,1,2,3]
Explanation: 40 is the largest element. 10 is the smallest. 20 is the second smallest. 30 is the third smallest.
```

Example 2:

```
Input: arr = [100,100,100]
Output: [1,1,1]
Explanation: Same elements share the same rank.
```

Example 3:

```
Input: arr = [37,12,28,9,100,56,80,5,12]
Output: [5,3,4,2,8,6,7,1,3]
```

Constraints:

- $0 <= arr.length <= 10^5$
- $-10^9 <= arr[i] <= 10^9$

For example:

Test	Result
print(arrayRankTransform([40,10,20,30]))	[4, 1, 2, 3]

Answer: (penalty regime: 0 %)

Reset answer

```python
def arrayRankTransform(arr: list[int]) -> list[int]:
    # Extract unique sorted elements
    sorted_unique = sorted(set(arr))

    # Map each unique value to its 1-based rank
    rank_map = {val: rank for rank, val in enumerate(sorted_unique, 1)}

    # Replace original elements with ranks
    return [rank_map[val] for val in arr]
```

	Test	Expected	Got	
✓	`print(arrayRankTransform([40,10,20,30]))`	`[4, 1, 2, 3]`	`[4, 1, 2, 3]`	✓
✓	`print(arrayRankTransform([100,100,100]))`	`[1, 1, 1]`	`[1, 1, 1]`	✓
✓	`print(arrayRankTransform([37,12,28,9,100,56,80,5,12]))`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	`[5, 3, 4, 2, 8, 6, 7, 1, 3]`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an integer array `nums`, return *an integer array* `counts` *where* `counts[i]` *is the number of smaller elements to the right of* `nums[i]`.

Example 1:

```
Input: nums = [5,2,6,1]
Output: [2,1,1,0]
Explanation:
To the right of 5 there are 2 smaller elements (2 and 1).
To the right of 2 there is only 1 smaller element (1).
To the right of 6 there is 1 smaller element (1).
To the right of 1 there is 0 smaller element.
```

Example 2:

```
Input: nums = [-1]
Output: [0]
```

Example 3:

```
Input: nums = [-1,-1]
Output: [0,0]
```

Constraints:

- `1 <= nums.length <= 10^5`
- `-10^4 <= nums[i] <= 10^4`

For example:

Test	Result
`print(countSmaller([5,2,6,1]))`	`[2, 1, 1, 0]`
`print(countSmaller([-1]))`	`[0]`

Answer: (penalty regime: 0 %)

Reset answer

```python
def countSmaller(nums: list[int]) -> list[int]:
    n = len(nums)
    counts = [0] * n
    # Store tuples of (value, original_index)
    enum_nums = list(enumerate(nums))

    def merge_sort(enum_arr):
        if len(enum_arr) <= 1:
            return enum_arr

        mid = len(enum_arr) // 2
        left = merge_sort(enum_arr[:mid])
        right = merge_sort(enum_arr[mid:])

        merged = []
        i = j = 0
        right_counter = 0  # Number of elements from right half smaller than left[i]

        while i < len(left) and j < len(right):
            # If left element is greater than right element
            if left[i][1] > right[j][1]:
                right_counter += 1
                merged.append(right[j])
                j += 1
            else:
                # Left element is smaller or equal; record accumulated smaller elements
                counts[left[i][0]] += right_counter
                merged.append(left[i])
                i += 1

        # Process remaining elements in left sub-array
        while i < len(left):
```

```
            counts[left[i][0]] += right_counter
            merged.append(left[i])
            i += 1

        # Append remaining elements in right sub-array
        while j < len(right):
            merged.append(right[j])
            j += 1

        return merged

    merge_sort([(i, num) for i, num in enumerate(nums)])
    return counts
```

	Test	Expected	Got	
✓	print(countSmaller([5,2,6,1]))	[2, 1, 1, 0]	[2, 1, 1, 0]	✓
✓	print(countSmaller([50,20,60,10]))	[2, 1, 1, 0]	[2, 1, 1, 0]	✓
✓	print(countSmaller([-1]))	[0]	[0]	✓
✓	print(countSmaller([-1, -1]))	[0, 0]	[0, 0]	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

To find the frequency of numbers in a list and display in sorted order.

Constraints:

1<=n, arr[i]<=100

Input:

1 68 79 4 90 68 1 4 5

output:

1 2

4 2

5 1

68 2

79 1

90 1

For example:

Input	Result
4 3 5 3 4 5	3 2 4 2 5 2

Answer: (penalty regime: 0 %)

```
import sys
from collections import Counter


def count_frequencies():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    # Parse inputs to integers
    nums = [int(x) for x in tokens]

    # Count frequency of each number
    freq = Counter(nums)

    # Output sorted by the numbers in ascending order
    for num in sorted(freq.keys()):
        print(f"{num} {freq[num]}")


if __name__ == "__main__":
    count_frequencies()
```

	Input	Expected	Got	
✓	4 3 5 3 4 5	3 2 4 2 5 2	3 2 4 2 5 2	✓
✓	12 4 4 4 2 3 5	2 1 3 1 4 3 5 1 12 1	2 1 3 1 4 3 5 1 12 1	✓

	Input	Expected	Got	
✓	5 4 5 4 6 5 7 3	3 1 4 2 5 3 6 1 7 1	3 1 4 2 5 3 6 1 7 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given an listof integers, sort the array in ascending order using the *Bubble Sort* algorithm above. Once sorted, print the following three lines:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

For example, given a worst-case but small array to sort: a=[6,4,1]. It took 3 swaps to sort the array. Output would be

```
Array is sorted in 3 swaps.
```

```
First Element: 1
```

```
Last Element: 6
```

Input Format

The first line contains an integer,n , the size of the list a .
The second line contains n, space-separated integers a[i].

Constraints

· 2<=n<=600

· $1<=a[i]<=2x10^6$.

Output Format

You must print the following three lines of output:

1. List is sorted in numSwaps swaps., where numSwaps is the number of swaps that took place.
2. First Element: firstElement, the *first* element in the sorted list.
3. Last Element: lastElement, the *last* element in the sorted list.

Sample Input 0

3

1 2 3

Sample Output 0

List is sorted in 0 swaps.

First Element: 1

Last Element: 3

For example:

Input	Result
3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3
5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9

Answer: (penalty regime: 0 %)

```
import sys


def count_swaps():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])
    a = [int(x) for x in tokens[1 : n + 1]]
```

```
    num_swaps = 0

    # Standard Bubble Sort algorithm
    for i in range(n):
        for j in range(n - 1):
            if a[j] > a[j + 1]:
                a[j], a[j + 1] = a[j + 1], a[j]
                num_swaps += 1

    print(f"List is sorted in {num_swaps} swaps.")
    print(f"First Element: {a[0]}")
    print(f"Last Element: {a[-1]}")


if __name__ == "__main__":
    count_swaps()
```

	Input	Expected	Got	
✓	3 3 2 1	List is sorted in 3 swaps. First Element: 1 Last Element: 3	List is sorted in 3 swaps. First Element: 1 Last Element: 3	✓
✓	5 1 9 2 8 4	List is sorted in 4 swaps. First Element: 1 Last Element: 9	List is sorted in 4 swaps. First Element: 1 Last Element: 9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums`, sort the array in **increasing** order based on the frequency of the values. If multiple values have the same frequency, sort them in **decreasing** order.

Print the *sorted array*.

Example 1:

```
Input:
6
1 1 2 2 2 3

Output:
3 1 1 2 2 2

Explanation: '3' has a frequency of 1, '1' has a frequency of 2, and '2' has a frequency of 3.
```

Example 2:

```
Input:
5
2 3 1 3 2
Output:
1 3 3 2 2
Explanation: '2' and '3' both have a frequency of 2, so they are sorted in decreasing order.
```

Example 3:

```
Input:
9
-1 1 -6 4 5 -6 1 4 1
Output:
5 -1 4 4 -6 -6 1 1 1
```

Constraints:

- `1 <= nums.length <= 100`
- `-100 <= nums[i] <= 100`

For example:

Input	Result
6 1 1 2 2 2 3	3 1 1 2 2 2
5 2 3 1 3 2	1 3 3 2 2

Answer: (penalty regime: 0 %)

```python
import sys
from collections import Counter


def frequency_sort():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])
    nums = [int(x) for x in tokens[1 : n + 1]]

    # Count frequencies of each element
    freq = Counter(nums)

    # Sort with custom key: (frequency ascending, value descending)
    nums.sort(key=lambda x: (freq[x], -x))

    print(" ".join(map(str, nums)))
```

```
20
21
22 if __name__ == "__main__":
23     frequency_sort()
```

	Input	Expected	Got	
✓	6 1 1 2 2 2 3	3 1 1 2 2 2	3 1 1 2 2 2	✓
✓	5 2 3 1 3 2	1 3 3 2 2	1 3 3 2 2	✓
✓	9 -1 1 -6 4 5 -6 1 4 1	5 -1 4 4 -6 -6 1 1 1	5 -1 4 4 -6 -6 1 1 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Bubble Sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements if they are in wrong order. You read an list of numbers. You need to arrange the elements in ascending order and print the result. The sorting should be done using bubble sort.

Input Format: The first line reads the number of elements in the array. The second line reads the array elements one by one.

Output Format: The output should be a sorted list.

For example:

Input	Result
6 3 4 8 7 1 2	1 2 3 4 7 8
5 4 5 2 3 1	1 2 3 4 5

Answer: (penalty regime: 0 %)

```
import sys


def bubble_sort():
    tokens = sys.stdin.read().split()
    if not tokens:
        return

    n = int(tokens[0])
    arr = [int(x) for x in tokens[1 : n + 1]]

    # Standard Bubble Sort algorithm
    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]

    print(" ".join(map(str, arr)))


if __name__ == "__main__":
    bubble_sort()
```

	Input	Expected	Got	
✓	6 3 4 8 7 1 2	1 2 3 4 7 8	1 2 3 4 7 8	✓
✓	6 9 18 1 3 4 6	1 3 4 6 9 18	1 3 4 6 9 18	✓
✓	5 4 5 2 3 1	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.